CHERGUELAINE Ayoub

AI / GenAI Engineer · LLMs · RAG · NLP · Data Engineering

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PROFESSIONAL SUMMARY

AI / GenAI Engineer with a Master's in Natural Language Processing, specializing in Large Language Models, Retrieval-Augmented Generation, agentic AI, and data engineering. I design and ship production-grade language systems — from fine-tuning open-source LLMs to building RAG and multi-agent pipelines — and back my work with publicly released fine-tuned models and datasets on Hugging Face. Comfortable across the full stack (FastAPI, Next.js) and cloud delivery (Docker, AWS), turning advanced AI research into reliable, scalable products.

SKILLS

Generative AI & LLMs: RAG, AI agents, LangChain, CrewAI, fine-tuning, embeddings, vector stores, prompt engineering, agent orchestration

ML & NLP: PyTorch, TensorFlow, Hugging Face, Transformers, scikit-learn, OCR, Arabic & dialect NLP, document intelligence

Data Engineering: ETL pipelines, scraping, cleaning, deduplication, validation, PostgreSQL, MongoDB, Redis, data visualization

Backend & Frontend: FastAPI, Flask, Next.js, React, TypeScript, Tailwind CSS, REST APIs

Cloud & DevOps: Docker, AWS (EC2, Lambda), Google Cloud, CI/CD, Git, workflow automation (n8n)

Languages: Python, TypeScript, JavaScript, SQL, Bash

WORK EXPERIENCE

Paseetah — AI & Data Engineer *Oct 2024 – Present*

Riyadh (Remote)

- Fine-tuned open-source LLMs into a custom classification agent, improving intent detection and property-query categorization.
- Designed and optimized ETL pipelines for large-scale real-estate and geospatial data (scraping, cleaning, deduplication, validation), processing **10M** records.
- Improved legacy dataset quality (parcels, transactions, roads) by cleaning and harmonizing **7M** records, raising data accuracy to **99%**.
- Applied ML, LLMs, OCR, and NLP to real-estate and Arabic data — transaction anomaly detection, Arabic text processing, translation, and information extraction.
- Built LLM agents to extract legal/regulatory rules from building codes, converting unstructured documents into structured, machine-readable data for automated compliance.

JURAI — LLM Engineer & Full-Stack Developer *Jan 2025 – Dec 2025*

Denmark (Remote) · Part-time

- Designed and implemented production-grade agentic AI systems using LangChain and CrewAI — retrieval, classification, reasoning, legal analysis, RAG, and decision-prediction agents.
- Developed the FastAPI backend and AI orchestration layer, integrating PostgreSQL/vector databases, external APIs, authentication, and scalable asynchronous workflows.
- Deployed, tested, and monitored AI services on AWS EC2 with Docker — production testing, logging, error handling, and fallback mechanisms.

BuySell — Front-End Developer *Feb 2024 – Dec 2024*

Blida, Algeria · Part-time

- Developed and improved admin-platform features with Next.js, TypeScript, and Tailwind CSS — dashboards, forms, tables, and management interfaces.
- Integrated and tested APIs across admin workflows and built core e-commerce features (product pages, category listings, search, filters, cart, navigation).

SIG Services — Python Developer *Sep 2023 – Dec 2023*

Blida, Algeria

- Designed and implemented Odoo ERP modules to meet business requirements, including API development and testing.

Icosnet — Artificial Intelligence Intern *Jan 2023 – Jul 2023*

Algiers, Algeria

- Built an intelligent, context-based document-classification system using transformer architectures, reaching **97%** accuracy on company documents.

SELECTEDPROJECTS

- DOCTool — Semantic OCR & Document AI — Built a full-stack Arabic and English document intelligence platform combining layout-aware OCR, structured extraction, hybrid RAG, and document agents.
- Mustashar — Data Pipelines & Regulation Agents — Developed automated data pipelines and specialist AI agents for Arabic regulations and salary data, integrating OCR, PostgreSQL, Neo4j, and source-cited answers.
- Jurai — AI Legal Assistant & Assurance — Built legal document analysis and bilingual flight-claim assessment applications combining OCR, precedent retrieval, case comparison, streaming LLM analysis, and validated legal citations.
- QueryLens — Conversational Data Analytics — Built an AI analytics workspace for CSV, SQLite, and PostgreSQL, enabling natural-language exploration through guarded read-only SQL, inspectable results, and downloadable reports.
- Company Documents v2 — Created a synthetic document generator and published 353,580 English and French business PDFs across 13 document types, with field annotations, word coordinates, and degraded scans.
- Saudi Arabic Dialect NLP Models — Developed and published fine-tuned models for Saudi Arabic dialect classification and conversational applications.
- LFM2.5-2.6B Fable5 Coding Agent — Fine-tuned a lightweight language model on agentic coding data for local coding assistance and instruction following.
- BART-MNLI CNN News Zero-Shot Classifier — Fine-tuned BART-Large-MNLI on CNN news and published the model and tokenizer for topic classification through the Hugging Face zero-shot pipeline.
- BERT Fine-Tuned on arXiv Metadata — Fine-tuned and published a BERT model for research-paper topic classification using arXiv metadata, with documented training and evaluation metrics.

EDUCATION

M.Sc., Natural Language Processing (Artificial Intelligence) *2021 – 2023*
Saad Dahleb Blida University, Algeria · GPA 13.0 / 20.0

B.Sc., Information Systems & Software Engineering *2018 – 2021*
Saad Dahleb Blida University, Algeria · GPA 11.0 / 20.0

CERTIFICATIONS

- Building Transformer-Based Natural Language Processing Applications — NVIDIA
- Fundamentals of Deep Learning — NVIDIA